Sanaul Haque

Sadia_Afrin_Faiza_Proposal

Course 608

Document Details

Submission ID
trn:oid:::3117:635601718

Submission Date
Oct 7, 2026, 10:39 PM GMT+6

Download Date
Oct 7, 2026, 10:40 PM GMT+6

File Name
Sadia_Afrin_Faiza_Proposal.pdf

File Size
368.0 KB

7 Pages

1,913 Words

10,812 Characters

11% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

- **3** Not Cited or Quoted 2%
  Matches with neither in-text citation nor quotation marks
- **4** Missing Quotations 2%
  Matches that are still very similar to source material
- **8** Missing Citation 8%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

10% Internet sources
9% Publications
7% Submitted works (Student Papers)

Match Groups

- **3** Not Cited or Quoted 2%
  Matches with neither in-text citation nor quotation marks
- **4** Missing Quotations 2%
  Matches that are still very similar to source material
- **8** Missing Citation 8%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

10% Internet sources
9% Publications
7% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Internet	www.dovepress.com 2%
2	Internet	uk.typology.com 2%
3	Internet	thesis.cust.edu.pk 2%
4	Publication	Malik Itrat, Sheema Suroor. "Therapeutic approaches to acne vulgaris in Unani m... 1%
5	Publication	Rony Aad, Simone Vesentini. "Spices", Elsevier BV, 2026 1%
6	Internet	jusst.org 1%
7	Internet	doaj.org <1%
8	Internet	mymoodmilk.com <1%
9	Internet	www.spandidos-publications.com <1%
10	Publication	Abejide Ade-Ibijola, Solomon Oyelere, Jude Osakwe. "Innovation in Africa for Indu... <1%

11 Publication

Shamama Javed, Muhammad H. Sultan, Waquar Ahsan, Andleeb Khan. "Dermatol... <1%

12 Publication

Magdalena Zielińska, Katarzyna Dereń, Ewelina Polak-Szczybyło, Agnieszka Ewa S... <1%

Research Proposal

On

Black Seed Oil as a Natural Alternative to 5 Synthetic Skincare Ingredients: A Comparative Theoretical Analysis.

Scholar Name:

Sadia Afrin Faiza

Name of Institution:

East West University

Date of Submission:

20/09/2026

International Academic Research Competition 2026

(Senior Level)

Black Seed Oil as a Natural Alternative to 5 Synthetic Skincare Ingredients:

A Comparative Theoretical Analysis.

Introduction :

8 The famous black seed oil comes from the seeds of Nigella Sativa , also known as black cumin . For hundreds of years people have been using black seed oil on their face . This black seed oil is deeply rooted into Islamic history. There is a hadith stating that black seed is the cure of everything except death. But now in this modern world the skincare also progress and use skincare made in laboratory which is mostly synthetic. As we go though the modern skincare ingredient we get to know that vitamin c protects skin from environmental damage and also provide a great antioxidant . Benzoyl peroxide kills the acne causing bacteria . Ceramides repairs the skin barrier and also provide moisturization. Peptides are basically amino acid bonds that provides great hydration and supports collagen production . and here comes the most studied ingredient that is Tretinoin and Tazarotene renew the skin [1]. These ingredients work without any doubt , but they often cause dryness ,redness , peeling , and irritation [1]. So here comes the black seed oil which can give similar benefit without too much side effects. Black seed oil have a main active compound which is known as Thymoquinone . the oil also contains of linoleic acid , oleic acid, and other useful compounds[2].

This study argues that access to basic skin care is not a luxury, but a public health necessity, especially for vulnerable populations who cannot afford or tolerate multiple synthetic products.

So in this research proposal I will compare the black seed oils compound with the five well know skincare ingredients to show that one natural oil can work as synthetic famous ingredients.

Research Questions :

1. Can black seed oil work like vitamin c to protect our skin from oxidative stress?
2. Does black seed oil help to clear up acne causing bacteria like Benzoyl Peroxide?
3. Can black seed oil help to moisturize and repair skin barrier like Ceramide?
4. Does blackseed oil have the ability to produce collagen in skin like Peptides?
5. Can black seed oil helps to renew skin cells like the infamous Tretinoin and Tazarotene?

Literature Review:

2 The antioxidant review : vitamin C is the most well known and researched antioxidant . Black seed oil also has strong antioxidant compounds. Thymoquinone enhances the activity of superoxide dismutase (SOD), glutathione peroxidase (GPx), and catalase enzymes that eliminate free radicals[2]. A review that N. sativa inhibits UV induced reactive oxygen species (ROS) and blocks the NF-kB pathway, which triggers inflammation and collagen breakdown[3].

Antibacterial effects: A 2025 randomized controlled trial compared a Nigella sativa formulation with 5% benzoyl peroxide gel in 40 participants with acne 28 days[4]. The trial suggested that the natural formulation may have therapeutic potential as an alternative acne treatment[4]. Another randomized double blind clinical trial using a N. sativa hydrogel found significant improvement in acne symptoms with acceptable tolerability[5].

Barrier repair: black seed oil is rich in linoleic acid. Linoleic acid reacts with ceramides in the outer skin layer through a process called esterification. This reaction helps form skins waterproof barrier and reduces water loss[2]. A deficiency in linoleic acid weakens the epidermal barrier and increase water loss, making skin drier and more prone to irritation[2].

Collagen buildup : A study using a standardized black cumin extract called Thymocid® found that it inhibited collagenase activity by 10.4% to 92.4% and elastase enzymes by 25.3% to 91.1%[6]. This means it protects collagen from breakdown. Thymoquinone also promotes collagen density and skin hydration, especially in skin exposed to sunlight[3].

Retinoid like effect: a clinical trial compared Nigella sativa cream(20%) with tretinoin cream(0.05 %) for acne vulgaris[7]. In a mouse tail model of psoriasis, N. sativa oil is safe compared to tretinoin and benzoyl peroxide, which cause local and systematic side effects[7].

Furthermore, studies on psoriasis models show that Nigella sativa extract increases relative epidermal thickness, confirming its ability to influence skin structure and support healthy cell renewal [8].

Theoretical Framework:

This is the core of the study. The framework is built on one simple idea. black seed oil contains several active compounds, and each compound can perform a job that a different synthetic ingredient normally does. Instead of five separate products, one oil may cover five functions.

Antioxidant Defense : Thymoquinone vs. Vitamin C

Vitamin C works by donating electrons to neutralise free radicals. Thymoquinone works differently. It enhances the skin's own protective enzymes SOD, GPx, and catalase [2]. This is a longer lasting defence because the skin builds its own protection. Thymoquinone also blocks NF-κB, a pathway that triggers inflammation and collagen breakdown after UV exposure [3]. So the antioxidant benefit comes with an extra anti-inflammatory benefit.

Antibacterial Action : Thymoquinone vs. Benzoyl Peroxide

Benzoyl peroxide kills Cutibacterium acnes by releasing oxygen. It works fast but irritates the skin. Black seed oil works differently. Thymoquinone and other compounds in the oil inhibit C. acnes and calm the

inflammation around acne lesions [5]. This means the oil attacks both the bacteria and the redness, which benzoyl peroxide does not do. The trial results support this similar acne reduction, but with better tolerability [4].

Barrier Repair: Linoleic Acid vs. Ceramides

Ceramides are lipids that hold the skin barrier together. Black seed oil does not contain ceramides, but it contains linoleic acid, which is the raw material the skin uses to build them [2]. Linoleic acid binds with ceramides through esterification to form the skin's impermeable barrier [2]. Black seed oil also contains oleic acid, which has a moisturizing effect and strengthens the hydrolipidic film [2]. So instead of adding ceramides from outside, black seed oil helps the skin build its own.

Collagen Support: Thymoquinone and Alpha-Hederin vs. Peptides

Peptides send signals telling the skin to make more collagen. Black seed oil works in two ways. First, it protects collagen that already exists. Thymocid® inhibited collagenase by up to 92.4% and elastase by up to 91.1% [6]. Second, it supports new collagen. Thymoquinone increases collagen density and improves skin hydration [3]. So black seed oil both shields and builds, while peptides mainly build.

Cell Renewal : Thymoquinone and Nigellamine vs. Tretinoin and Tazarotene

Tretinoin and Tazarotene work by binding to retinoic acid receptors, which speeds up skin cell turnover. This works well but causes peeling and redness [1]. Nigella sativa extract produced epidermal differentiation effects similar to tazarotene in a psoriasis model [8]. This suggests black seed oil influences the same cell renewal pathways, but more gently. The head to head trial of N. sativa cream versus tretinoin cream is testing exactly this idea [7].

Studies on psoriasis models also show that Nigella sativa extract increases relative epidermal thickness, confirming its ability to influence skin structure and support healthy cell renewal [8].

The framework summary:

Synthetic Ingredient	The work its done by	Black seed oil compound	How it works
Vitamin C	Antioxidant	Thymoquinone	Enchances SOD,GPx ,catalase.
Benzoyl peroxide	Kills acne bacteria	Thymoquinone	Inhibits c. acnes, calms inflammation.
Ceramides	Repairs barrier	Linoleic acid, Oleic acid	Builds acylceramide, seals water loss[2].
Peptides	Supports collagen	Thymoquinone ,Alpha-hederin	Blocks collagenase and elastase, boosts collagen density[6],[3].
Tretinoin/ Tazarotene	Renew skin cells	Thymoquinone	Promotes epidermal differentiation and also increases epidermal thickness .[7],[8].

Proposed Methodology::

This research proposal is literature based study. I approach this proposal by using qualitative method . In this proposal I did not use any participants or experiments. It compares existing science based evidence. I choose this method because I am an individual researcher. I do not have a laboratory . In quantitative studies we need equipment to measure skin hydration, water loss, and enzyme activity[6]. These are not accessible to me. A literature review is low cost and practical. I just complete my proposal with bringing all the studies in one frame.

Expected outcome :

I expected to show that black seed oil can work like most famous active ingredients on skin. I expected that the thymoquinone in black seed will be the strongest anti oxidant and antibacterial because of the clinical studies available[4],[5]. And the weakest was one skin barrier and skin renewal because less study has conducted on this . And I wanted to show that black seed oil causes less problems than benzoyl peroxide and tretinoin.

Potential Limitations

This study relies on published literature, so it depends on the quality of those studies. Some trials are small. Some use different formulations and different concentrations of black seed oil, which make

comparison difficult. Black seed oil composition also varies by source and extraction method. Since this is a literature based study, the findings will be theoretical and will need future experimental testing.

Conclusion:

This project is a practical pilot. It brings together existing scientific evidence to test one idea. that black seed oil can do the work of five synthetic skincare ingredients. The theoretical framework is the heart of the study. If the evidence supports the framework, it could help develop affordable, plant based skincare for people who cannot tolerate synthetic products. The findings will be shared with dermatology clinics, skincare formulators, and community health organizations.

Ultimately, this research advocates for the view that taking care of our skin the body's largest organ is fundamental necessity that should be accessible to everyone, regardless of the

References:

[1] A. Evanti, T. L. Wargasetia, and J. W. Gunadi, "Role of *Nigella sativa* L. seed (black cumin) in preventing photoaging (Review)," *Biomed. Rep.*, vol. 23, no. 2, **p. 131**, Jun. 2025, doi: 10.3892/br.2025.2009.

[2] Typology, "Topical Application: What Are the Benefits of Black Seed Oil?," Typology, June 19, 2023. Available: us.typology.com/library/topical-application-what-are-the-benefits-of-black-seed-oil. [Accessed: September 21, 2026].

[3] N. S. Alrawi et al., "Thymoquinone and its protective role against oxidative stress in skin," Antioxidants, vol. 11, no. 5, p. 942, May 2022. Available: mdpi.com. [Accessed: September 21, 2026].

[4] S. N. A. Abbas *et al.*, "Evaluation of the efficacy of topical *Nigella sativa* L. with vinegar in the treatment of acne vulgaris: A randomized controlled trial," *J. Dermatol. Sci. Cosmet. Technol.*, vol. 2, no. 1, **p. 100069**, Mar. 2025.

[5] S. Soleymani et al., "The effect of a hydrogel made by Nigella sativa L. on acne vulgaris: A randomized double-blind clinical trial," Phytother. Res., vol. 34, no. 11, pp. 3052–3062, Nov. 2020, doi: 10.1002/ptr.6739.

[6] H. Li *et al.*, "Thymocid®, a Standardized Black Cumin (*Nigella sativa*) Seed Extract, Modulates Collagen Cross-Linking, Collagenase and Elastase Activities, and Melanogenesis in Murine B16F10 Melanoma Cells," *Nutrients*, vol. 12, no. 7, **p. 2146**, Jul. 2020, doi: 10.3390/nu12072146.

[7] Islamic Azad University, "Efficacy and safety of Nigella sativa cream (20%) vs. tretinoin cream (0.05%) for acne vulgaris: a randomized right/left double-blind comparative clinical trial," Iranian Clinical Trials Registry, Trial ID IRCT20191021045173N1, p. 1, Oct. 2019. Available: irct.behdasht.gov.ir. [Accessed: September 21, 2026].

[8] S. C. Dwarampudi et al., "Antipsoriatic activity and cytotoxicity of ethanolic extract of Nigella sativa seeds," Pharmacogn. Mag., vol. 8, no. 32, pp. 268–272, Oct.–Dec. 2012; compiled in "Recent Advances in Psoriasis Therapy," Bentham Science Publishers, Table 1, 2023. Available: benthamscience.com. [Accessed: September 21, 2026].